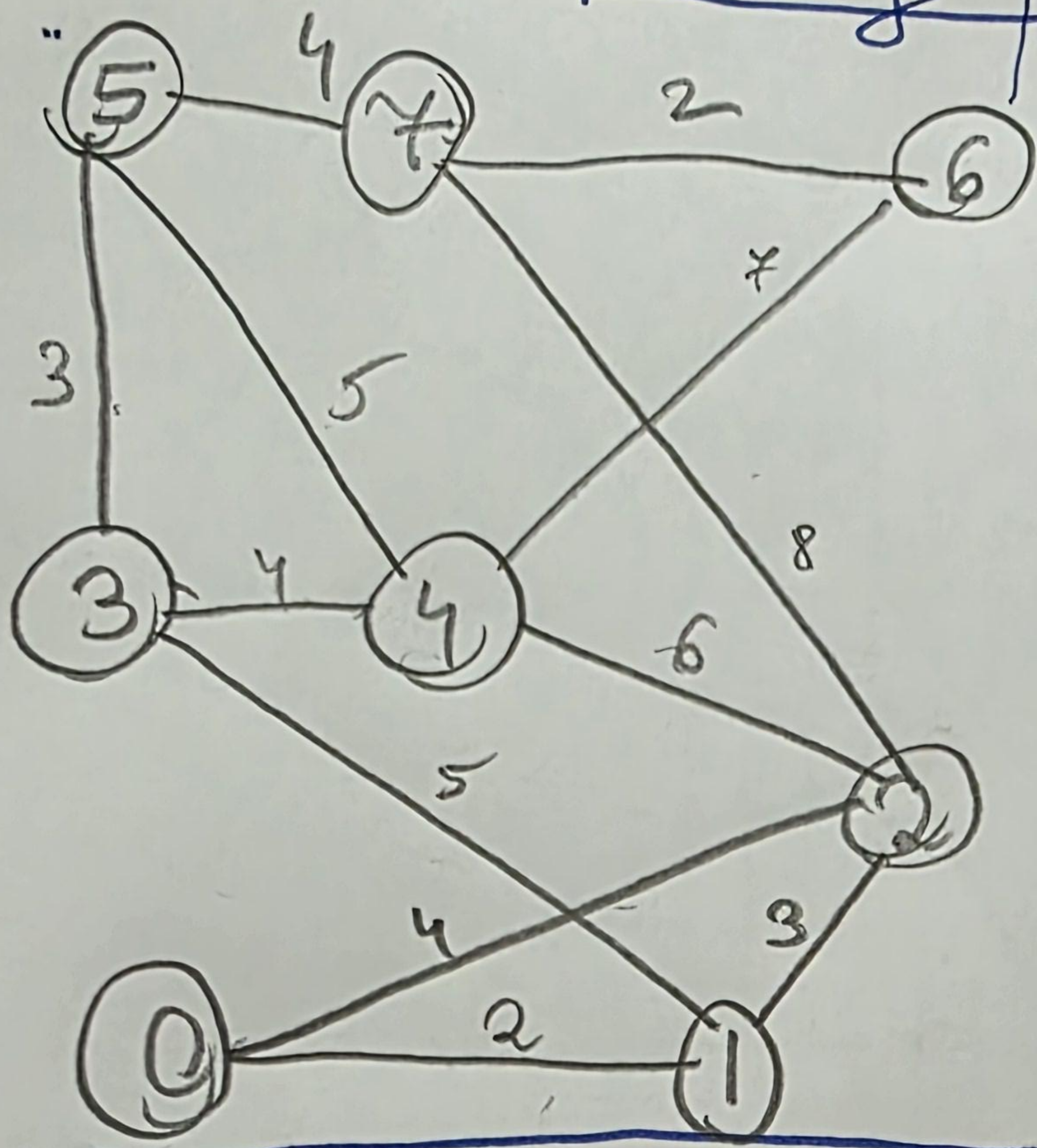


Minimum spanning tree



Edges

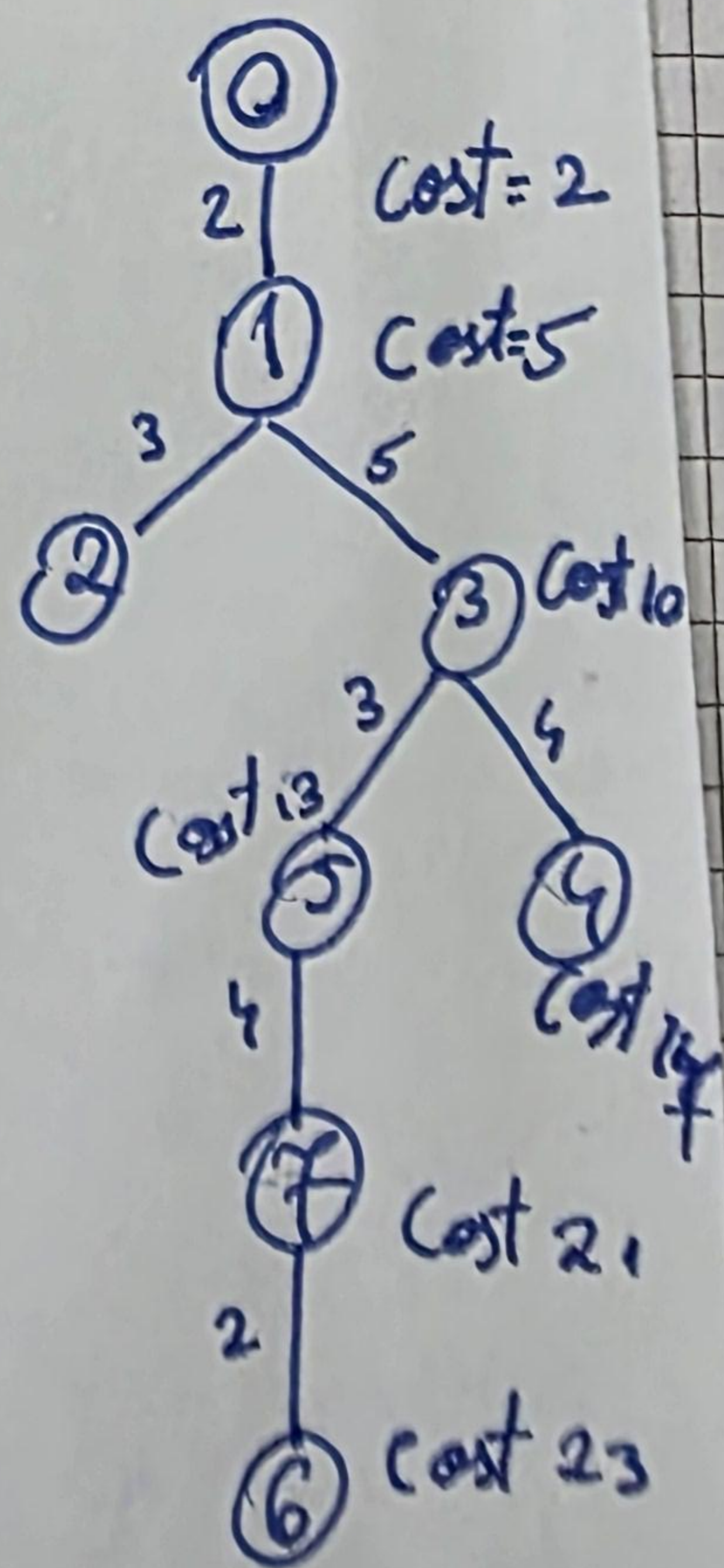
- (0,1), 2
- (0,2), 4
- (1,2), 3
- (1,3), 4
- (2,3), 5
- (2,4), 6
- (3,4), 4
- (3,5), 3
- (4,5), 5
- (4,6), 7
- (5,7), 4
- (6,7), 2
- (2,7), 8

Edges sorted

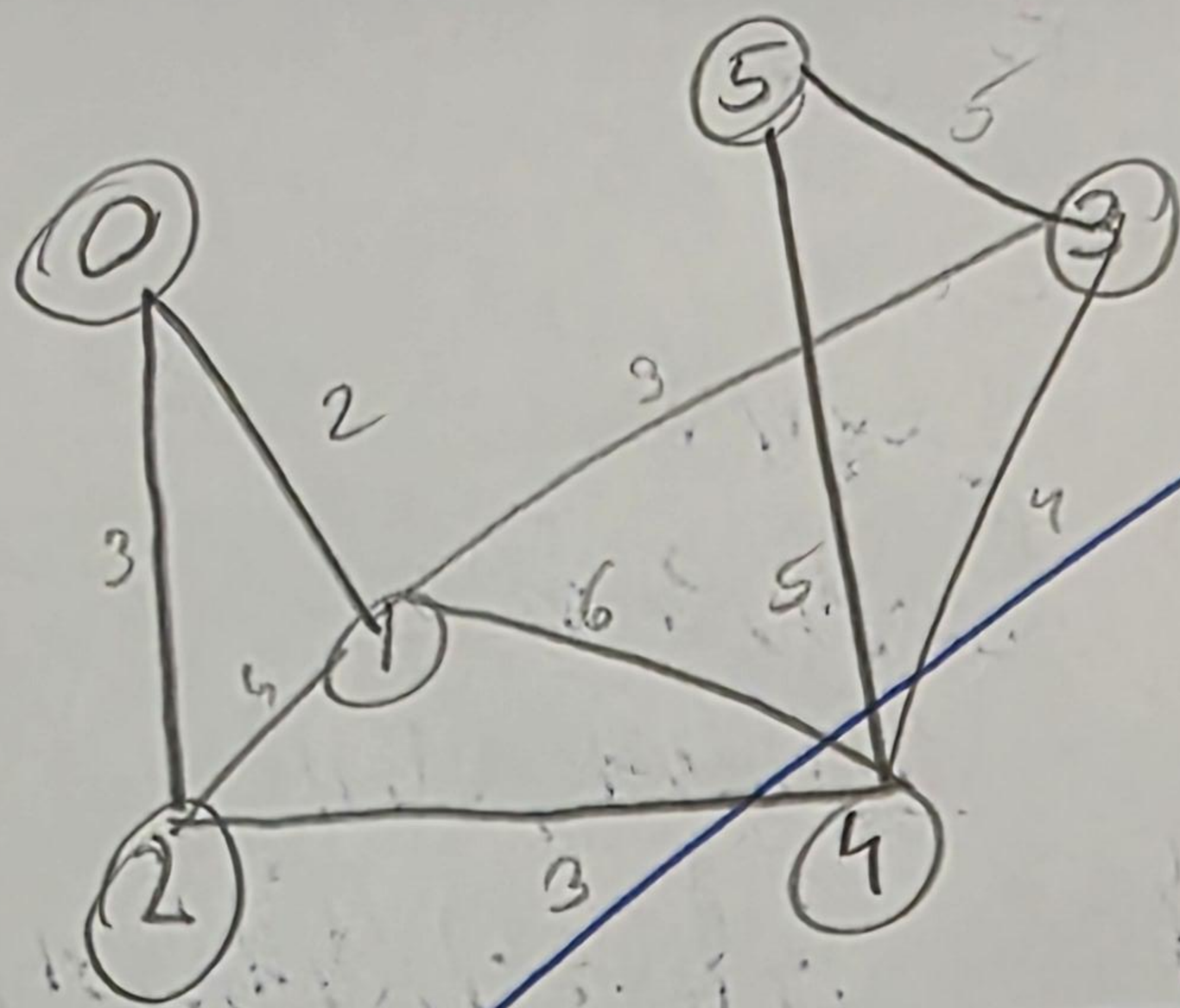
- (0,1) - 2
- (6,7) - 2
- (1,2) - 3
- (3,5) - 3
- (0,2) - 4
- (3,4) - 4
- (5,7) - 4
- (1,3) - 5
- (4,5) - 5
- (2,4) - 6
- (4,6) - 7
- (2,7) - 8

Prim's algorithm

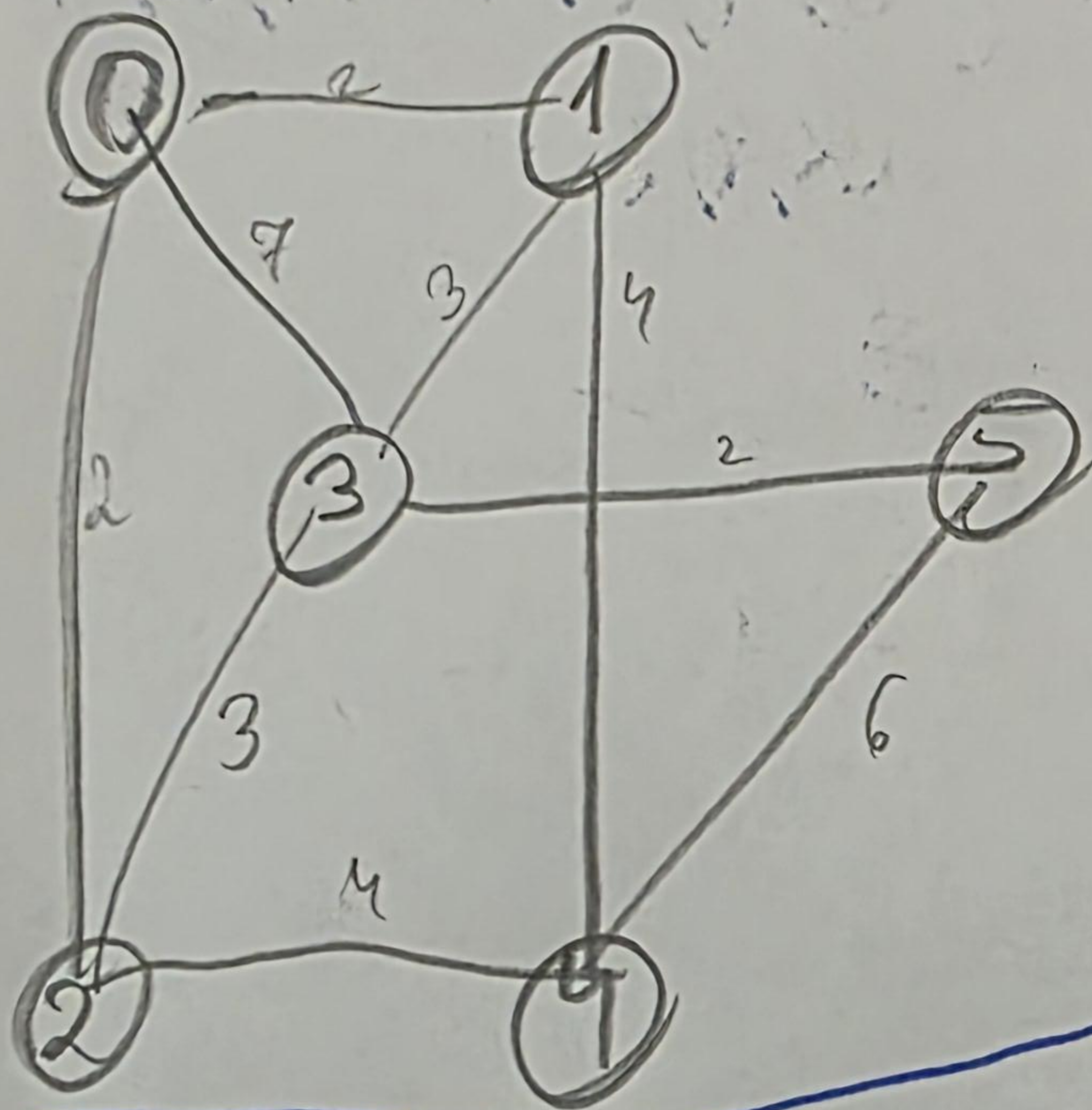
	selected edge	V_{new}	E_{new}	MST
initialization		$\{0\}$	$\{\}$	
iteration 1	(0,1)	$\{0,1\}$	$\{(0,1)\}$	
iteration 2	(1,2)	$\{0,1,2\}$	$\{(0,1), (1,2)\}$	
iteration 3	(1,3)	$\{0,1,2,3\}$	$\{(0,1), (1,2), (1,3)\}$	
iteration 4	(3,5)	$\{0,1,2,3,5\}$	$\{(0,1), (1,2), (1,3), (3,5)\}$	
iteration 5	(3,4)	$\{0,1,2,3,4,5\}$	$\{(0,1), (1,2), (1,3), (3,5), (3,4)\}$	
iteration 6	(5,7)	$\{0,1,2,3,4,5,7\}$	$\{(0,1), (1,2), (1,3), (3,5), (3,4), (5,7)\}$	
iteration 7	(7,6)	$\{0,1,2,3,4,5,6,7\}$	$\{(0,1), (1,2), (1,3), (3,5), (3,4), (5,7), (7,6)\}$	



Minimum spanning tree



Edges	Edges
(0,1) - 2	(0,1) - 2
(0,2) - 3	(0,2) - 3
(1,2) - 4	(1,3) - 3
(1,3) - 3	(2,4) - 3
(2,4) - 3	(1,2) - 4
(3,4) - 4	(3,4) - 4
(3,5) - 5	(3,5) - 5
(4,5) - 5	(4,5) - 5
(1,4) - 6	(1,4) - 6



(0,1) - 2	(0,1) - 2
(0,2) - 2	(0,2) - 2
(1,3) - 3	(3,5) - 2
(2,3) - 3	(1,3) - 3
(1,4) - 4	(3,4) - 3
(2,4) - 4	(1,4) - 4
(3,5) - 2	(2,4) - 4
(4,5) - 6	(4,5) - 6
(0,3) - 7	(0,3) - 7

MST1

Init

sel. edge

V_{new}

E_{new}

It. 1

(0,1)

$\{0,1\}$

$\{(0,1)\}$

It. 2

(0,2)

$\{0,1,2\}$

$\{(0,1), (0,2)\}$

It. 3

(1,3)

$\{0,1,2,3\}$

$\{(0,1), (0,2), (1,3)\}$

It. 4

(3,5)

$\{0,1,2,3,5\}$

$\{(0,1), (0,2), (1,3), (3,5)\}$

It. 5

(1,4)

$\{0,1,2,3,4,5\}$

$\{(0,1), (0,2), (1,3), (3,5), (1,4)\}$

$$\text{Total} = 2 + 2 + 3 + 2 + 4 = 13$$

MST 2

	Sel. edge	V_{new}	E_{new}
Init	—	$\{0\}$	$\{ \}$
It. 1	$(0,1)$	$\{0,1\}$	$\{(0,1)\}$
It. 2	$(0,2)$	$\{0,1,2\}$	$\{(0,1), (0,2)\}$
It. 3	$(1,3)$	$\{0,1,2,3\}$	$\{(0,1), (0,2), (1,3)\}$
It. 4	$(3,5)$	$\{0,1,2,3,5\}$	$\{(0,1), (0,2), (1,3), (3,5)\}$
It. 5	$(2,4)$	$\{0,1,2,3,4,5\}$	$\{(0,1), (0,2), (1,3), (3,5), (2,4)\}$

MST 3

	Sel. edge	V_{new}	E_{new}
Init	—	$\{0\}$	$\{ \}$
It. 1	$(0,1)$	$0,1$	
It. 2	$(0,2)$	$0,1,2$	
It. 3	$(0,3)$	$0,1,2,3$	
It. 4	$(3,5)$	$0,1,2,3,5$	
It. 5	$(1,4)$	$0,1,2,3,4,5$	

All cost 13

